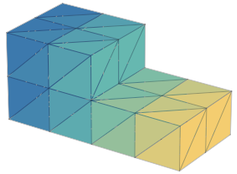
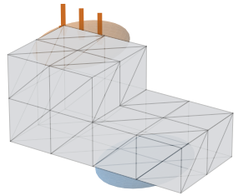
1 Spatial physical hypotheses



Material field



Load + support fields

Material contrast; unknown load / support parameters

2 Integrate unknown loads

Possible force histories



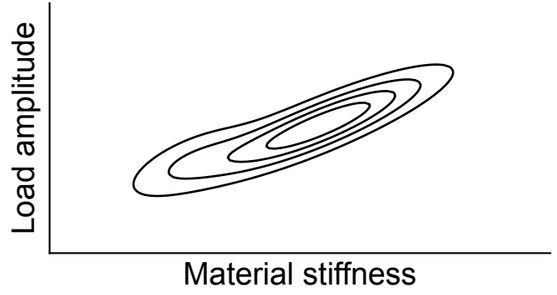
$$y = H(\theta, \beta)\, F + \varepsilon$$

$$F \sim N(0, K)$$

$$C = HKH^T + R$$

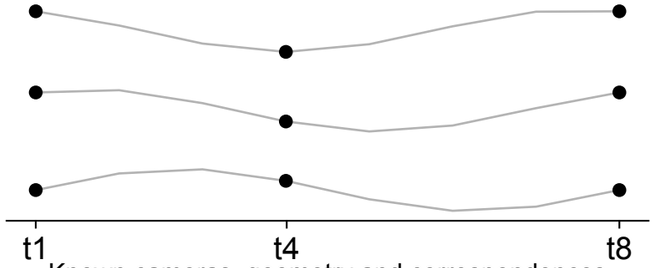
Exact conditional Gaussian integral

3 Retain joint uncertainty



Preserve alternative coupled explanations

Observed displacement tracks

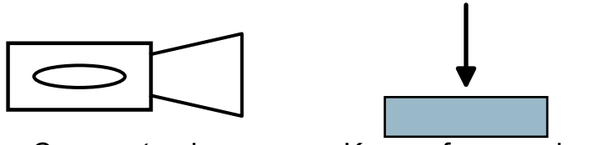


Known cameras, geometry and correspondences

new evidence

4 Look or probe?

Choose the smallest expected prediction risk



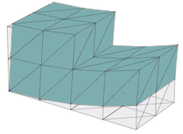
Camera track

Known-force probe

Same target, same scalar measurement budget

5 Predict held responses

Hidden deformation + changed loading



Report prediction error and interval coverage

Method schematic. Mesh geometry is real; fields, curves and deformation are illustrative, not experimental results.